

TED (10) – 5075

Reg. No.....

REVISION – 2010

Signature.....

MODEL QUESTION PAPER

(Common for EI & IT)

AIRCRAFT INSTRUMENTS

(Maximum Marks: 100)

(Time: 3 hours)

PART – A

(Answer all questions in one or two sentences. Each question carries two marks) Marks

- I
1. Interpret flight instruments.
 2. State critical Mach number.
 3. List the two units of atmospheric pressure.
 4. State Precession.
 5. List two Temperature sensors used in Aircraft. (5×2 =10)

PART – B

(Answer any five questions. Each question carries six marks)

- II
1. Summarize the requirements of the pitot static instruments.
 2. Differentiate between quantitative and qualitative displays.
 3. Explain any one method of heating circuit arrangement for pitot probe.
 4. Explain the methods of operating Gyroscopic flight Instruments.
 5. Explain with Diagram Linear and Non-linear Scales.
 6. Explain Trace Recording.
 7. Explain Electromagnetic Recording. (5×6 = 30)

Marks

PART – C

(Answer one full question from each unit. Each question carries fifteen marks)

UNIT - I

- III 1. Explain the gear mechanism in indicating instruments. 8
2. Explain the temperature compensation method used in aircraft instruments. 7

OR

- IV 1. Describe what type of display would you associate with the following
 Instruments. a) Altimeter b) gyroscope 8
2. Explain straight scale displays in aircraft compare it with circular displays. 7

UNIT – II

- V 1. With a neat diagram, explain the working of Mach meter. 8
2. Describe with diagram the working of Aneroid barometer. 7

OR

- VI 1. With the help of a diagram, explain the sensing and transmission of pitot
 pressure to the instruments. 8
2. Derive the equation for $P = \frac{1}{2} \rho V^2$. 7

UNIT – III

- VII 1. Explain the working of Generator and Indicator System 8
2. Describe Gyro Horizon. 7

OR

- VIII 1. With a Diagram, explain the working of DC Torque motor Tachometer. 8
2. Explain the working of Tacho Probe 7

UNIT – IV

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| IX | 1. Explain the working of LVDT Pressure Transmitter. | 8 |
| | 2. Describe the principle and operation of Thermocouple. | 7 |

OR

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| X | 1. Explain the working of Basic Capacitance Fuel Gauge system. | 8 |
| | 2. Describe the principle and operation of Basic Accelerometer. | 7 |
